

BAAZ JKBOSE



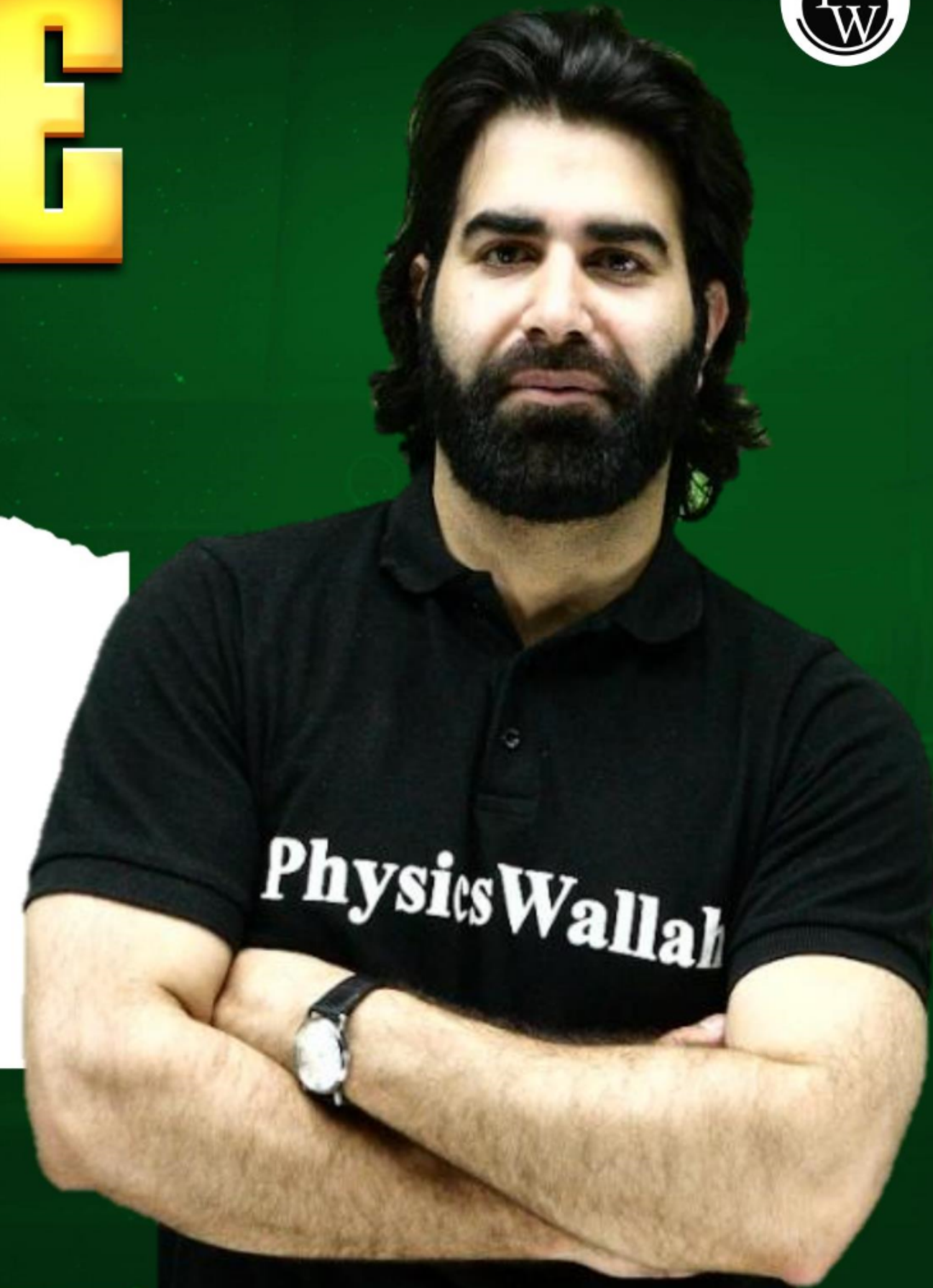
2026 CLASS-11TH

Breathing and Gaseous Exchange

LECTURE-5

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Transport of gases
- 2 

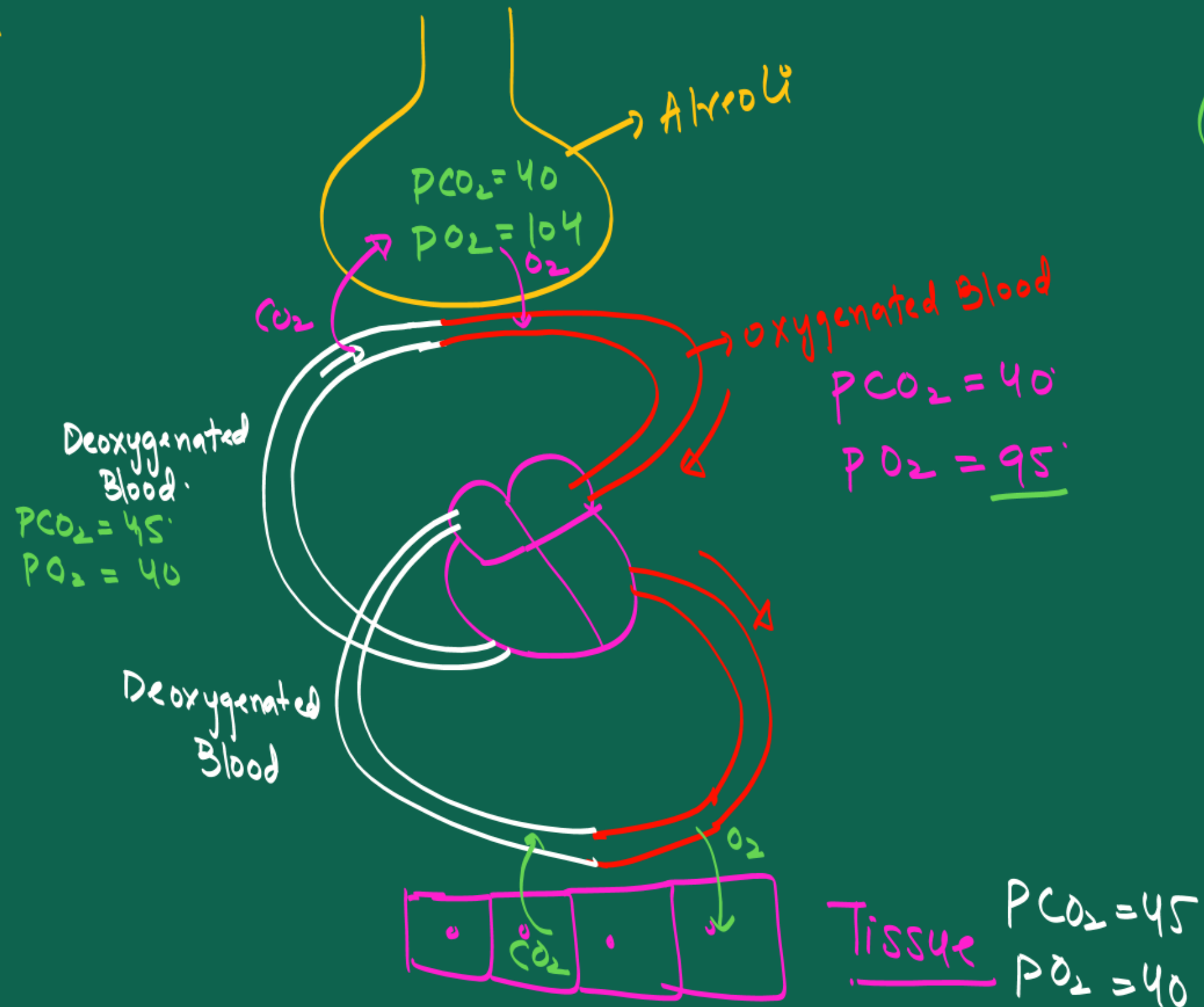
Revision

CO_2 A D O T
 40 45 40 45
 104 40 95 40

	Atmosphere	Alveoli	Deoxy Blood	Oxy Blood	Tissues
PO_2	159	104	40	A=95	40
PCO_2	0.3	40	45	40	45

A D O T.
 PCO_2 40 45 40 45
 104 40 95 40

⇒ Gaseous Exchange



Pressure = mmHg

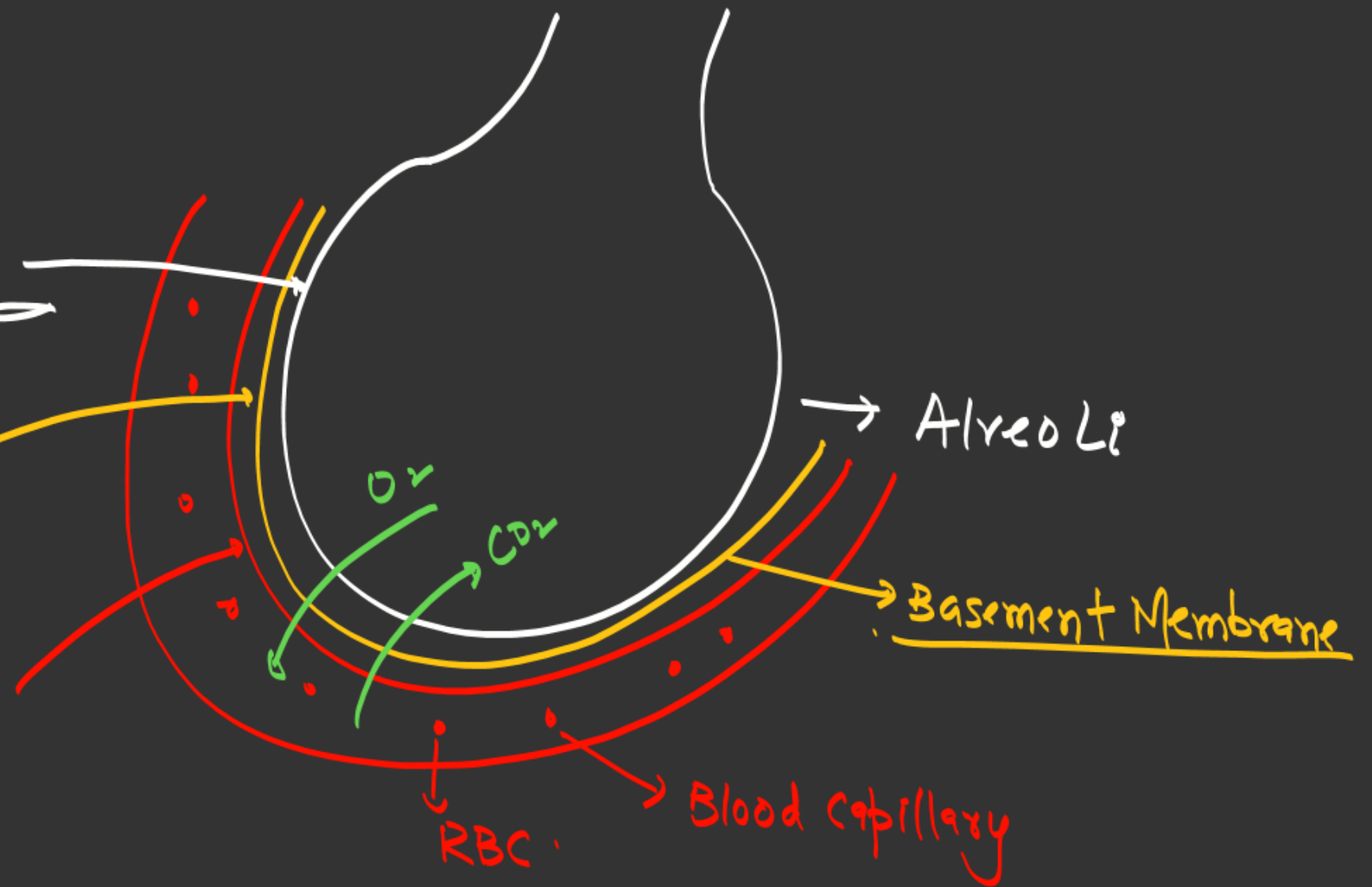
Diffusion Membrane at Lungs.

→ It is Made of Three Layers

① Simple Squamous of Alveoli

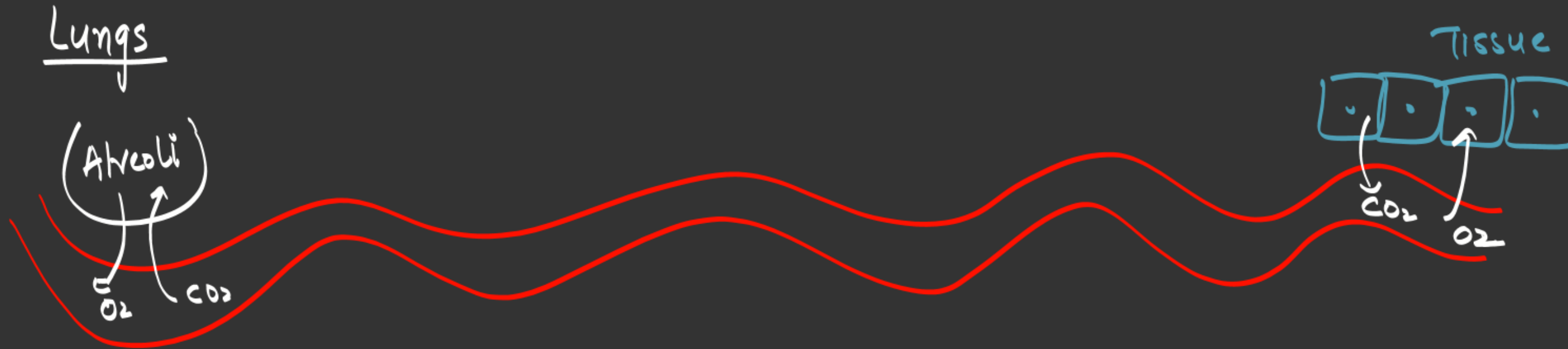
② Basement Membrane (Non-cellular permeable).

③ Simple Squamous (Endothelium)
of Blood Capillary.



Transport of Gases

→ In this topic we will discuss, How gases are transported from tissue to lungs / Lungs to tissue.



→ We will discuss;

① Transport of Oxygen:

② Transport of Carbon.

Transport of oxygen.

- oxygen is carried in two ways

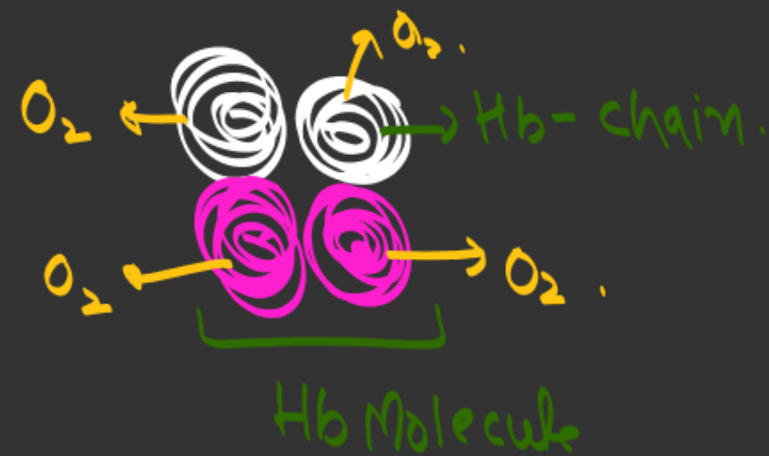
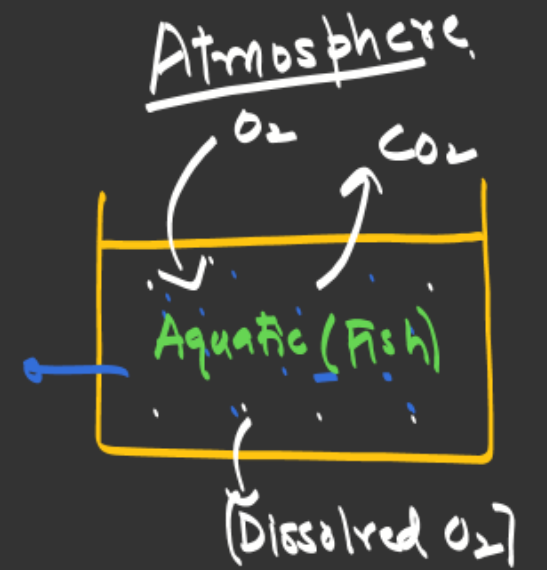
① Dissolved oxygen - It is in the plasma (liquid) of our blood.
- It carries 3% of total oxygen.

② Oxy-Haemoglobin - Oxygen is mainly carried as Oxy-Haemoglobin.

[Oxygen + Haemoglobin] - 97%.

- A Haemoglobin Molecule is made of 4 Hb chain
- To one Hb chain, 1 molecule of O_2 can bind.

→ Adult ($\alpha_2 \beta_2$)
 H_2O



∴ To one Hb, Molecule, 4 O_2 Can Bind.

NCERT.

100ml of Blood, can deliver (delivers) 5ml of O_2 to Tissues under Normal Conditions.

Explanation

→ At an Average, an Adult has 15gm of Hb in 100ml of Blood.

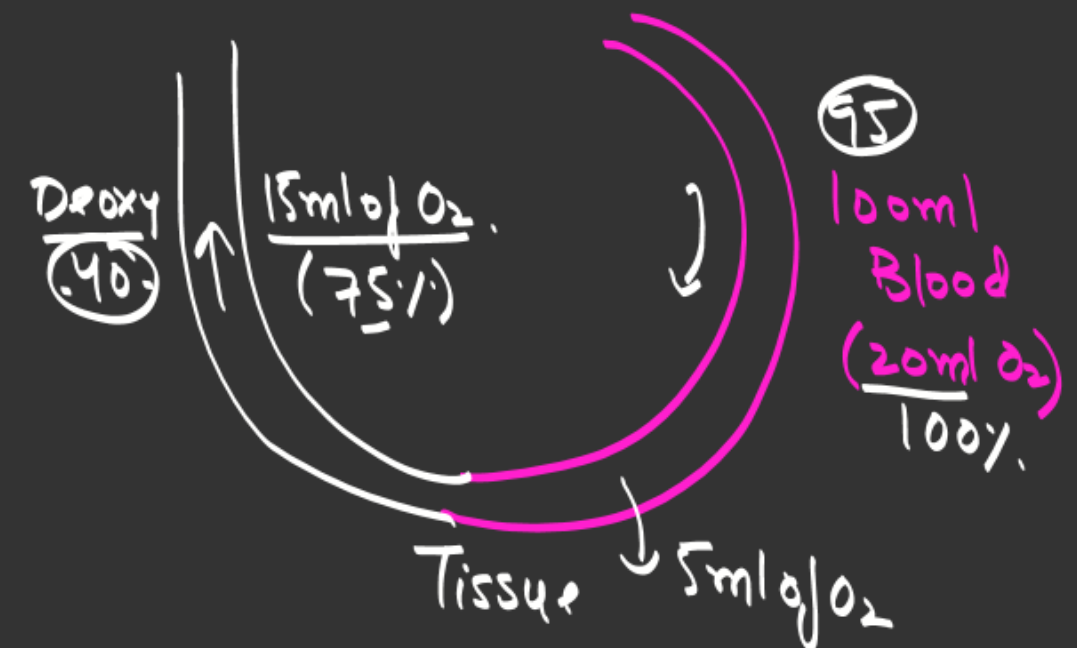
→ One gm of Hb. carries 1.34ml of O_2 .

→ 15gm of Hb will carry $\frac{15 \times 1.34}{(\approx 20)}$ ml of O_2 .

→ We can Also say,

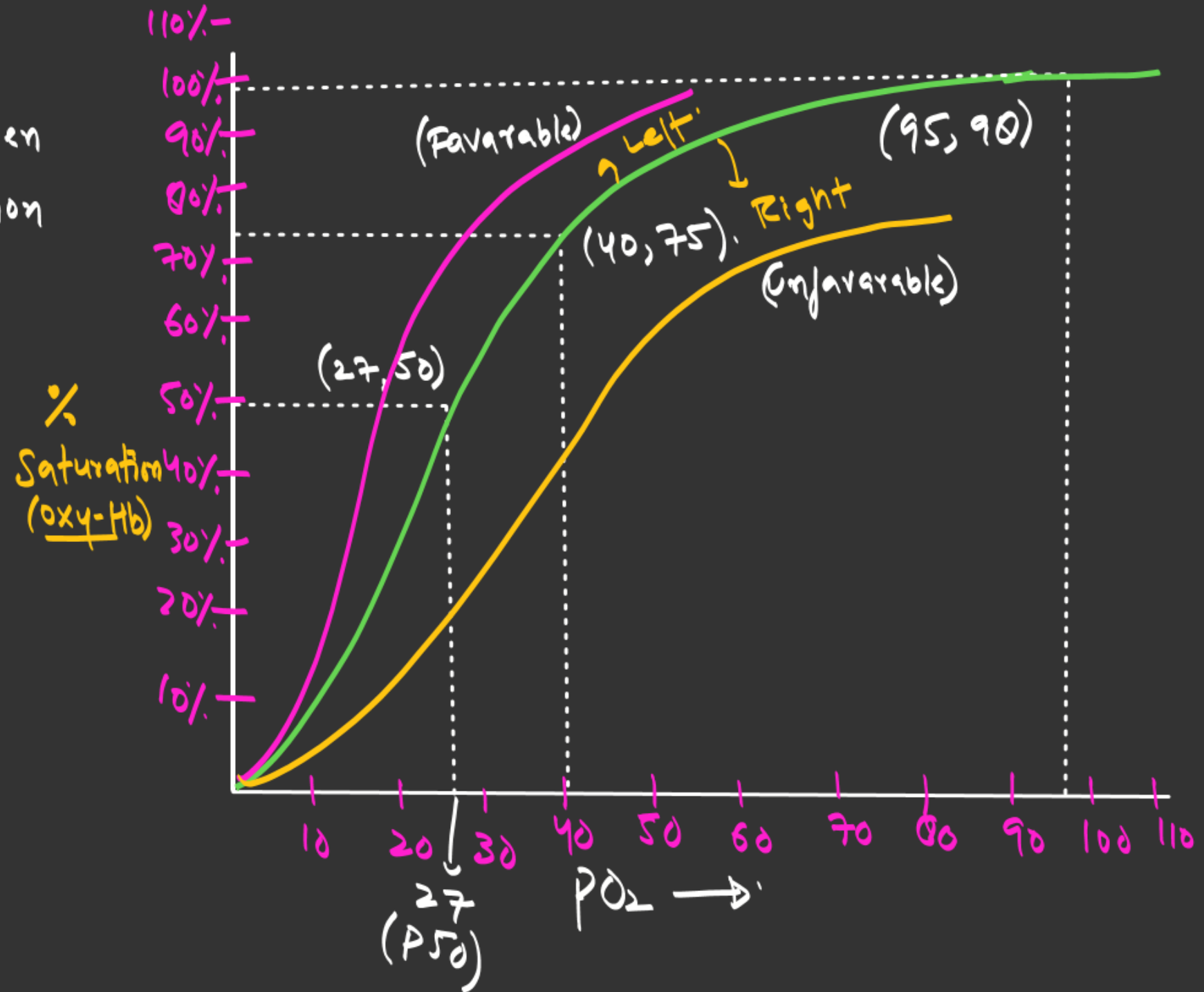
100ml of Blood carries 20ml of O_2

During High Exercise
100ml of Blood
delivers 15ml of
 O_2 .



Oxygen Haemoglobin Dissociation Curve

- This is also known as Oxy-Hb Curve
- It is a graphical Representation between PO_2 (partial pressure of oxygen) and % Saturation of Oxy-Hb
- We get S-Shaped (Sigmoid) Curve.



Factors Affecting oxy-Hb Curve

There are Five factors,

- ① PO_2 ② PCO_2 ③ H^+ Conc ④ Temp ⑤ BPG.
(Bisphosphoglycerate)

- Formed in RBC:

Favorable conditions for oxy-Hb formation

$PO_2 \uparrow$ → these conditions are at lungs.

$PCO_2 \downarrow$

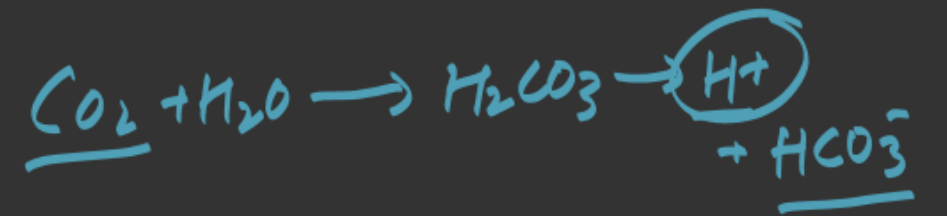
(Increased pH) $H^+ \downarrow$

(oxy-Hb)

Temp $\downarrow$

BPG $\downarrow$

→ This shifts curve to Left



Unfavorable condition for oxy-Hb formation.

$PO_2 \downarrow$

- These conditions are at Tissue

$PCO_2 \uparrow$

(Decreased pH) $H^+ \uparrow$

(oxy-Hb)

Temp $\uparrow$

BPG $\uparrow$

→ This shifts curve to Right

QUESTIONS



The partial pressure of oxygen is equal in

- 1 Atmospheric air $\frac{159}{159}$ and Alveoli $\frac{104}{104}$
- 2 Alveoli $\frac{104}{104}$ and Oxygenated blood $\frac{95}{95}$
- 3 Alveoli $\frac{104}{104}$ and Deoxygenated blood $\frac{40}{40}$
- 4 Deoxygenated blood $\frac{40}{40}$ and Tissues $\frac{40}{40}$

QUESTIONS



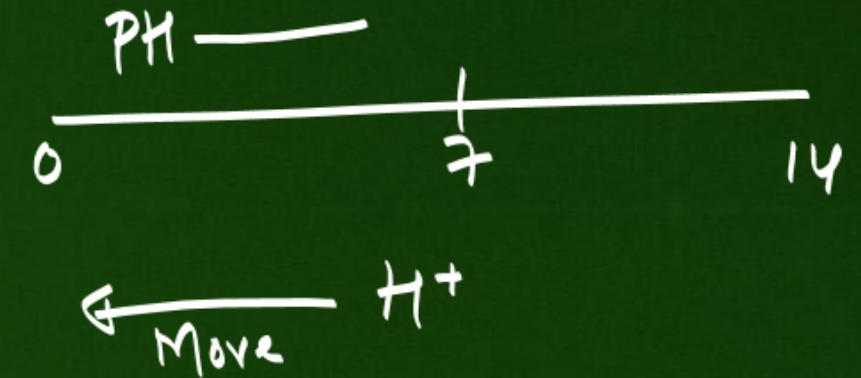
Which of the following does not shift the oxy-haemoglobin dissociation curve to the right? (Unfavorable)

1 Increased pH $\uparrow H^+ \times$ $pH \propto \frac{1}{H^+}$
 $H^+ \downarrow$

2 Increased carbon dioxide $PCO_2 \uparrow$

3 Increased temperature $Temp \uparrow$

4 Increased 2,3-dpg BPG



Thank
You